PERFECT AND EXTREME BINARY HUMBERT FORMS OVER REAL QUADRATIC FIELDS IN THE MULTIPLICATIVE NORMALISATION: AN ENUMERATION FOR 157 FIELDS, PATTERNS, AND NEXT STEPS

ABSTRACT. For every real quadratic field $K = \mathbb{Q}(\sqrt{D})$, $D \leq 300$ squarefree, whose Hermite–Humbert constant $\gamma_{K,2}$ (Icaza's multiplicative normalisation) has been certified in this project — 157 of the 182 fields — we list all classes of perfect binary Humbert forms modulo $\mathrm{GL}_2(\mathcal{O}_K)$ and scaling, and decide eutaxy for each. There are 55444 perfect classes, of which 15313 are extreme, 80 weakly eutactic and 40051 not eutactic; in every field the largest γ of a perfect form equals the certified $\gamma_{K,2}$. We describe the enumeration (a branch-and-bound over the fundamental region with certified discarding rules, followed by a local Newton solve), prove that it is complete under one explicitly stated numerical hypothesis, and discuss how to remove that hypothesis. The data suggest three regularities: the number of perfect classes is proportional to $\zeta_K(-1)$, i.e. to the volume of the Hilbert modular surface, with no regulator factor (in contrast with the additive normalisation); $\gamma_{K,2}$ is described to within about 20% (residual standard deviation 0.18 in log) by $d_K^{1.17}\mathrm{Reg}_K^{-0.61}$, and we prove the lower bound $\gamma_{K,2} \geq 4\pi^2\zeta_K(-1)/(\sqrt{d_K}\,\mathrm{Reg}_K)$ for $h_K = 1$; and forms with an automorphism of order 3 have $\frac{3}{4}\gamma$ equal to a norm from $\mathbb{Z}[\zeta_3]$, reflecting a CM structure. Every claim is labelled [PROVED], [CERTIFIED] (certified or exhaustive computation, range stated) or [CONJECTURE].

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Date: Working notes, November 2026.

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1. Introduction

Let $K = \mathbb{Q}(\sqrt{D})$ be a real quadratic field with discriminant $d = d_K$, ring of integers $\mathcal{O}_K$, fundamental unit $\varepsilon_0 > 1$, regulator $\mathrm{Reg}_K = \log \varepsilon_0$ and class number $h = h_K$. A *binary Humbert form* is a pair $S = (S_1, S_2)$ of positive definite real 2×2 matrices, S_j attached to the real embedding $x \mapsto x^{(j)}$. Following Icaza and [1, 8, 9] we use the *multiplicative* normalisation (1)

$$S[v] = S_1[v^{(1)}]\, S_2[v^{(2)}], \quad m(S) = \min_{0 \neq v \in \mathcal{O}_K^2} S[v], \quad \gamma(S) = \frac{m(S)}{\sqrt{\det S_1 \det S_2}}, \quad \gamma_{K,2} = \sup_S \gamma(S).$$

This is a different theory from the *additive* normalisation $S_1[v^{(1)}] + S_2[v^{(2)}]$ of Koecher, Leibak and Gunnells–Yasaki [5, 6, 4]: the minimum is a product rather than a sum, so it is not a linear function of S, and there is no Ryshkov polyhedron. The two are never mixed below; the additive theory is the subject of the companion project `pav10/perfect_forms`, which we only quote for comparison.

The constant $\gamma_{K,2}$ has been determined for small discriminants in [1, 8, 9]; in this project it was certified for 157 fields with $D \leq 300$ (branch-and-bound over the fundamental region, every certificate re-checked with outward rounding; see the project README). The present notes report a second computation on top of the first: the complete list of *perfect* forms (in Coulangeon's sense [3]) and their *eutaxy*, for all 157 fields. The extreme forms (perfect and eutactic) are the local maxima of γ; the global maximum is $\gamma_{K,2}$.

Summary.

(1) *Algorithm* (§3). Voronoi's neighbour algorithm is not available in the multiplicative setting (§3.1). We enumerate instead: a branch-and-bound over the region of highest points discards boxes with three certified rules, and the remaining small boxes are solved locally. Theorem 3.4 proves completeness under a single hypothesis (H) on the local Newton solve, and §3.5 explains how interval Newton would remove it. The cost is proportional to $\zeta_K(-1)$ (§3.4).
(2) *Results* (§4). 55444 perfect classes in 157 fields (Table 1); validation against the literature, against the certified $\gamma_{K,2}$, and by an independent re-run at half the box size.
(3) *Patterns* (§5). A volume law $n_{\mathrm{perf}} \approx 7.9\, \zeta_K(-1)$ (Conjecture 5.2); an empirical law and a proved lower bound for $\gamma_{K,2}$ (Theorem 5.4); CM structure of the symmetric forms (Observation 5.8).
(4) *Next steps* (§6) and the literature we need (§7).

Conventions. Labels: [PROVED] — proved here or in a cited source; [CERTIFIED] — established by a certified or exhaustive computation in the stated range (for the enumeration of perfect forms this means: complete under hypothesis (H) of §3.3); [CONJECTURE] — conjecture or heuristic. Fits are least-squares fits of logarithms; R^2 is the coefficient of determination. All numbers are produced by `analysis/patterns.py` from `results/perfect/` and `results/gamma_K2.tsv`.

2. Setting

2.1. **Forms as points of $\mathbb{H}^2$.** Up to the two independent scalings $(S_1, S_2) \mapsto (\lambda_1 S_1, \lambda_2 S_2)$, which leave γ unchanged, a form is a point $z = (z_1, z_2) \in \mathbb{H}^2$, $z_j = x_j + iy_j$, via $S_j \propto \frac{1}{y_j} \begin{pmatrix} 1 & x_j \\ x_j & |z_j|^2 \end{pmatrix}$. For $v = (a, b) \in \mathcal{O}_K^2$,

$$f_v(z) = \frac{|a^{(1)} + b^{(1)} z_1|^2\, |a^{(2)} + b^{(2)} z_2|^2}{y_1 y_2}, \qquad G(z) = \min_{v \neq 0} f_v(z) = \gamma(S).$$

G is invariant under $\mathrm{GL}_2(\mathcal{O}_K)$ and under the Galois swap $(z_1, z_2) \mapsto (z_2, z_1)$, and $f_{\varepsilon v} = f_v$ for units ε, so minimal vectors come in *unit classes*. The space of forms modulo scaling is thus $\mathbb{H}^2$ itself (real dimension 4), and forms modulo equivalence are points of the Hilbert modular surface $\mathrm{PGL}_2(\mathcal{O}_K)\backslash\mathbb{H}^2$. (In the additive normalisation only the common scaling is free, and the space has dimension 5; see §5.1.)

2.2. **Highest points and the search region.** For $h = 1$ every orbit has a *highest* point (maximal y_1y_2), and there $e_1 = (1, 0)$ is a minimal vector, so $G(z) = f_{e_1}(z) = 1/(y_1y_2)$ [1, Prop. 1.1]. Using translations by $\mathcal{O}_K$, $z \mapsto \varepsilon_0^k z$, $x \mapsto -x$ (all in $\mathrm{PGL}_2(\mathcal{O}_K)$) and the Galois swap, every orbit has a highest point in

$$\mathcal{R} = \{x = r_1 + r_2\omega,\ r_1 \in [-\tfrac{1}{2}, \tfrac{1}{2}],\ r_2 \in [0, \tfrac{1}{2}];\ y_2/y_1 \in [1, \varepsilon_0];\ y_1y_2 \geq 2/d\},$$

$\omega = (1 + \sqrt{D})/2$ or $\sqrt{D}$; the last condition is Cohn's bound $\gamma_{K,2} \leq d/2$ [2]. For $h > 1$ the project uses one *chart* per pair of ideal classes $\{C, C^{-1}\}$: in the chart of an integral ideal $\mathfrak{c}$ of minimal norm in C the lattice is $\mathfrak{c}^{-1} \oplus \mathfrak{c}$, with the same region up to the translation lattice (project README, "Steinitz charts"). We write $t = \frac{1}{2}\log(y_1y_2)$, $s = \frac{1}{2}\log(y_2/y_1)$, and use box coordinates (x_1, x_2, s, t); within a chart put $N_v = f_v/f_{e_1}$, so that $N_v(z) = |a^{(1)} + b^{(1)}z_1|^2|a^{(2)} + b^{(2)}z_2|^2$.

2.3. **Perfection and eutaxy.** For a form S and a minimal vector v let

$$X_v = \big(S_2[v^{(2)}]\, v^{(1)}v^{(1)T},\ S_1[v^{(1)}]\, v^{(2)}v^{(2)T}\big),$$

the gradient of $S \mapsto S[v]$ in the 6-dimensional space $\mathrm{Sym}_2(\mathbb{R})^2$; $X_{\varepsilon v} = X_v$ for units. S is *perfect* if the X_v (v minimal) span a space of dimension 5, the maximum possible [3, 1]; *eutactic* if $(S_1^{-1}, S_2^{-1}) = \sum_v \rho_v X_v$ with all $\rho_v > 0$; *weakly eutactic* if this holds with $\rho_v \geq 0$ but not with $\rho_v > 0$; and *extreme* (a strict local maximum of γ) iff perfect and eutactic [3].

3. The enumeration

3.1. **Why not Voronoi's algorithm.** In the additive theory $S \mapsto S_1[v^{(1)}] + S_2[v^{(2)}]$ is linear, the forms with minimum ≥ 1 form a locally finite convex polyhedron (Ryshkov), perfect forms are its vertices, and Voronoi's algorithm walks from vertex to neighbouring vertex [10, 5, 4]. In the multiplicative theory the constraint $S[v] \geq 1$ is not linear (it is $\log S_1[v^{(1)}] + \log S_2[v^{(2)}] \geq 0$), the region $\{m \geq 1\}$ is not convex, and the local structure at a perfect point — although described by Coulangeon's theory — does not give a finite combinatorial "neighbour" step that is guaranteed to succeed. We implemented such a walk (`voronoi.py`: Lipschitz-certified steps along the edges of the arrangement $\{f_v = G\}$); it works on generic vertices but fails on degenerate faces, where more than the expected number of f_v meet. We therefore enumerate.

3.2. **The algorithm.** Fix a chart. Let $\mathcal{V}$ be a finite set of unit classes containing every class v with $f_v \leq d/2$ somewhere on $\mathcal{R}$ (it exists and is computed exactly as in the $\gamma_{K,2}$-certificate). For a box B of (x_1, x_2, s, t)-space with $t \leq 0$:

Rule 3: discard B if it misses the translation parallelogram;
Rule 1: discard B if $\sup_B N_v < 1$ for some $v \in \mathcal{V}$ (then e_1 is not minimal anywhere in B, so $B \cap \mathcal{R}' = \emptyset$, $\mathcal{R}'$ the set of points of $\mathcal{R}$ at which e_1 is minimal);
Rule 5: discard B if fewer than 5 classes $v \in \mathcal{V}$ (including e_1, for which $N_{e_1} \equiv 1$) satisfy $\inf_B N_v \leq 1$.

The bounds $\sup_B N_v$, $\inf_B N_v$ are the product bounds of the $\gamma_{K,2}$-certificate (exact interval bounds for $|a + bz|^2$ on a box, evaluated in floating point with a relative safety margin 10^{-9}). Boxes that survive are bisected along the coordinate of largest *hyperbolic* width ($\Delta x_j/y_j$, Δs, Δt) until that width is $\leq w_{\min} = 0.05$. In a surviving leaf B, with candidate set $C_B = \{v \neq e_1 : \inf_B N_v \leq 1\}$:

(1) for every 4-subset $\{v_1, \ldots, v_4\} \subset C_B$, run damped Newton on $\log N_{v_i}(q) = 0$, $i = 1, \ldots, 4$, in $q = (x_1, x_2, \log y_1, \log y_2)$, from the centre of B and three random points of B; accept the first limit that lies in B (with a margin of 1% of the box size);

(2) polish an accepted point to 50 digits; compute the full set M of minimal classes *independently of* $\mathcal{V}$ by the exact short-vector enumeration; reject the point if e_1 is not minimal; compute the rank of $\{X_v\}_{v\in M}$ and keep the point if it is 5;
(3) decide eutaxy by a linear programme (and, when $|M| = 5$, by solving for the unique ρ at 50 digits).

Finally the perfect points from all charts are grouped into $\mathrm{GL}_2(\mathcal{O}_K)$-classes by [8, Alg. 11] in exact arithmetic, and the stabiliser of each class in $\mathrm{GL}_2(\mathcal{O}_K)/\mathcal{O}_K^\times$ is computed. Large fundamental units are handled by cutting the s-range into slabs.

3.3. **Correctness.**

Lemma 3.1 ([PROVED])**.** *Let $z \in \mathcal{R}'$ be perfect. Then z has a minimal class $v = (a, b)$ with $b \neq 0$, and $y_1y_2 \leq 1$, i.e. $t \leq 0$; in the chart of $\mathfrak{c}$, $y_1y_2 \leq N(\mathfrak{c})^{-1} \leq 1$. For $h = 1$ every minimal class other than that of e_1 has $b \neq 0$.*

Proof. For a minimal $v = (a, 0)$ we have $S_j[v^{(j)}] = a^{(j)2}S_j[e_1]$ and $S[v] = S[e_1] = m$, so $X_v = \left(\frac{m}{S_1[e_1]}E_{11}, \frac{m}{S_2[e_1]}E_{11}\right) = X_{e_1}$ ($E_{11} = e_1e_1^T$). So classes with $b = 0$ add nothing to the rank, and a perfect point (rank 5) has a minimal class with $b \neq 0$. For it, $f_v \geq (b^{(1)}y_1)^2(b^{(2)}y_2)^2/(y_1y_2) = N(b)^2y_1y_2$, while $f_v = f_{e_1} = 1/(y_1y_2)$; hence $(y_1y_2)^2 \leq N(b)^{-2}$. For $h = 1$, $|N(b)| \geq 1$; in a chart $b \in \mathfrak{c}$, so $|N(b)| \geq N(\mathfrak{c}) \geq 1$. Finally, for $h = 1$ and $b = 0$, $f_v = N(a)^2 f_{e_1}$ forces $|N(a)| = 1$, so a is a unit and v is in the class of e_1. □

Lemma 3.2 ([PROVED])**.** *At $z \in \mathcal{R}'$ with minimal classes $M \ni e_1$, the rank of $\{X_v\}_{v\in M}$ is 5 if and only if the differentials $d\log N_v$, $v \in M \setminus \{e_1\}$, span the 4-dimensional cotangent space of $\mathbb{H}^2$ at z. In that case some 4 of them are linearly independent, and z is an isolated, nondegenerate solution of the corresponding 4×4 system $\log N_{v_i} = 0$.*

Proof. For minimal v, $\langle X_v, (S_1, 0)\rangle = \langle X_v, (0, S_2)\rangle = S[v] = m$, so all X_v lie in an affine subspace A of codimension 2 not containing 0. Hence $\operatorname{rank}\{X_v\} = 1 + \operatorname{rank}\{X_v - X_{e_1}\}$, and the differences lie in the annihilator of the two scaling directions, i.e. in the cotangent space of the quotient $\mathbb{H}^2$, of dimension 4. There $X_v - X_{e_1}$ is m times $d\log(S[v]/S[e_1]) = d\log N_v$. So rank 5 $\iff$ the $d\log N_v$ span a space of dimension 4, and the Jacobian of a suitable 4-subset is invertible at z. □

Lemma 3.3 ([PROVED], up to floating point)**.** *Rules 1, 3 and 5 never discard a box containing a perfect point $z \in \mathcal{R}'$ with $t \leq 0$. Every minimal class of such a z lies in $\mathcal{V}$ and in the candidate set of every leaf containing z.*

Proof. At z, $N_v(z) \geq 1$ for all v (e_1 minimal), so $\sup_B N_v \geq 1$: rule 1 does not apply; z lies in the parallelogram: rule 3 does not apply; z has at least 5 minimal classes (perfection needs rank 5), each with $N_v(z) = 1$, hence $\inf_B N_v \leq 1$: rule 5 does not apply. A minimal class has $f_v(z) = G(z) \leq \gamma_{K,2} \leq d/2$, so it is in $\mathcal{V}$. The bounds are exact interval bounds evaluated in floating point with a safety margin; "up to floating point" refers to this evaluation, which is the same as in the $\gamma_{K,2}$-certificate (there re-checked with outward rounding). □

Hypothesis (H). *For every perfect point z and every leaf $B \ni z$, Newton for at least one 4-subset of minimal classes of z with invertible Jacobian (Lemma 3.2) converges to z from one of the four starting points, and no other solution in B of the same subset is reached first.*

Theorem 3.4 (completeness, [PROVED] under (H))**.** *Under (H) the enumeration returns a representative of every $\mathrm{GL}_2(\mathcal{O}_K)$-class of perfect binary Humbert forms, and only perfect forms.*

Proof. Every class has a highest point in some chart's region with e_1 minimal and, by Lemma 3.1, $t \leq 0$. The leaves and discarded boxes tile the box cover of the region, and by Lemma 3.3 z lies in a leaf B whose candidate set contains its minimal classes. By (H) some 4-subset produces z; the polished point has the minimal set and rank computed independently of $\mathcal{V}$, so it is kept, and only points with rank 5 are kept. The classification is exact. □

Remark 3.5 (what is not rigorous). (i) Hypothesis (H): Newton has a nonsingular solution in the box but convergence from the chosen starts is not proved; the solve is local (box width ≤ 0.05 in hyperbolic terms) and started from 4 points, and a failure would need all of them to fail. (ii) The rank 5 and the eutaxy decisions are made at 50 digits with tolerances (10^{-25} relative for singular values, 10^{-9} for the eutaxy margin), so a weakly eutactic class is weakly eutactic to within 10^{-9}. (iii) The class list is checked against the certified $\gamma_{K,2}$ only through its maximum.

3.4. **Cost.** Over the 157 fields the search used $7.2 \cdot 10^8$ boxes, $5.3 \cdot 10^7$ leaves and $1.5 \cdot 10^9$ Newton runs. The number of boxes grows like $\zeta_K(-1)^{1.02}$ ($R^2 = 0.91$, the 127 fields with $d \geq 100$), the same law as the number of perfect classes (§5.1): the work is proportional to the volume of the Hilbert modular surface, i.e. roughly to $d^{3/2}$. The single most expensive field took 3.2 hours on one core ($D = 263$). Each class is found 3.1 times on average (different charts, translates and slabs), and 7.2% of the classes only once.

3.5. **Making it robust and faster.**

(1) *Certified completeness.* Replace (H) by an interval Newton (Krawczyk) test per 4-subset and leaf: either it proves that the subset has no solution in B, or that it has exactly one; leaves where neither holds are bisected. At a perfect point the Jacobian of some subset is invertible (Lemma 3.2), so near it the test succeeds after finitely many bisections; degenerate (non-perfect) crossings are rejected by the rank test. Together with an interval check of the rank and of the eutaxy LP this makes the lists certified.
(2) *Fewer subsets.* Most of the time goes to 4-subsets with no common zero in the leaf. Pairwise and triple tests (interval bounds on $\log N_v - \log N_w$) discard most of them before Newton.
(3) *Fewer copies.* Restricting the region by the stabilisers of the cusps (and the chart symmetries) would remove most of the 3.1 copies per class.
(4) *Parallelism inside a field.* Boxes are independent; splitting the s- or t-range over processes removes the long tail (now one process per field).
(5) *A compiled kernel* for the box bounds and the batched Newton. (The batched Newton already gave a factor ≈ 3 over the per-subset loop.)
(6) *A multiplicative neighbour walk* with exact handling of degenerate faces would make the cost proportional to the output; a mass formula over the cells (as in the additive census of `perfect_forms`) would then give an independent completeness check.

3.6. **Is the algorithm new?** The ingredients are known: highest points and the fundamental-domain search [1, 8, 9, 2], perfection and eutaxy in this setting [3], the equivalence test [8, Alg. 11]. To our knowledge the combination — certified discarding rules that isolate all candidates for perfect points, followed by an exhaustive local solve and an exact classification — and the resulting complete lists of multiplicative perfect forms for more than a handful of fields are new. This needs a literature check (§7): in particular we have not seen M. Wagner's thesis **[REF?]** or later work of Pohst–Wagner, Icaza or Coulangeon on enumerations.

4. Results

Table 1. Totals over the 157 fields with certified $\gamma_{K,2}$, $D \leq 300$ [certified].

perfect classes	55444
extreme (perfect + eutactic)	15313
weakly eutactic (in 52 fields)	80
not eutactic	40051
fields with one perfect class	$D = 2, 3, 5$
most perfect classes	$D = 274$: 1467
median fraction extreme	0.332

4.1. **Validation.**

- [CERTIFIED] In every one of the 157 fields the largest γ of a perfect class equals the certified $\gamma_{K,2}$ (relative difference $\leq 10^{-12}$).
- [CERTIFIED] The test suite reproduces the literature: one class for $D = 2$ and $D = 5$ [1] ($|\mathrm{Aut}| = 10$ for $D = 5$), two classes for $D = 13$, both extreme, the critical one with $\gamma = (84 + 6\sqrt{91})/35$; and $D = 10$ (7 perfect, 5 extreme). Where the experimental neighbour walk completed, it found the same classes.
- [CERTIFIED] Independent re-run: the fields with $D \leq 77$ listed in the file `robustness.txt` of `results/perfect/` were recomputed with half the leaf width ($w_{\min} = 0.025$, hence different leaves and Newton starts); the class lists (value of γ to 25 digits, number of minimal classes, eutaxy, $|\mathrm{Aut}|$) agree in every field.
- An audit after the short-vector fix (project history): one record changed ($D = 46$ had a spurious duplicate class with one minimal vector missed; now 84 perfect, 39 extreme).

4.2. **The table.** Table 2 in the appendix lists n_{perf} and n_{ext} for every field; the file `D⟨D⟩.json` in `results/perfect/` contains every class (value of γ, closed form when PSLQ finds one, minimal vectors, rank, eutactic coefficients, $|\mathrm{Aut}|$, a point).

5. Patterns

5.1. **A volume law for the number of perfect forms.**

Observation 5.1 ([CERTIFIED], 157 fields). Over the 127 fields with $d \geq 100$ the log-log slope of n_{perf} against $\zeta_K(-1)$ is 1.017 ($R^2 = 0.984$); against $\zeta_K(-1)\mathrm{Reg}_K$ it is 0.807 ($R^2 = 0.892$), against d it is 1.525 ($R^2 = 0.973$), against $h\mathrm{Reg}_K$ 1.604 ($R^2 = 0.54$). For the 98 fields with $d > 200$, $n_{\mathrm{perf}}/\zeta_K(-1)$ has median 7.90, quartiles 7.37, 8.56 and range $[5.96, 10.86]$; the mass $\sum 1/|\mathrm{Aut}|$ behaves the same way (median 7.74). For the extreme forms the slope against $\zeta_K(-1)$ is 0.823.

Conjecture 5.2 ([CONJECTURE]). *$n_{\mathrm{perf}}(K) \asymp \zeta_K(-1)$ as $d_K \to \infty$, with constants independent of Reg_K and h_K.*

Heuristic. By Hirzebruch's formula **[REF?]** the orbifold Euler characteristic of $\mathrm{PSL}_2(\mathcal{O}_K)\backslash\mathbb{H}^2$ is $2\zeta_K(-1)$, and its hyperbolic volume (measure $\prod_j dx_j\, dy_j/y_j^2$) is $8\pi^2\zeta_K(-1)$; $\zeta_K(-1) \asymp d^{3/2}L(2, \chi_d)$. In the multiplicative normalisation the space of forms modulo scaling is exactly $\mathbb{H}^2$, so a locally finite configuration of perfect points with bounded local density has $\asymp$ vol classes. In the additive normalisation the separate scalings of S_1 and S_2 are not symmetries; the space has one more dimension, the log-unit direction, and `perfect_forms` (FINDINGS §3, additive, $d \leq 100$) finds $n \asymp \zeta_K(-1)\mathrm{Reg}_K$ there; for unary forms Yasaki's bound [11] relates the count to the period of a continued fraction, i.e. to Reg_K. The three laws are consistent with "number of perfect forms $\asymp$ volume of the space of forms modulo $\mathrm{GL}_n(\mathcal{O}_K)$ and the symmetries of the normalisation".

The constant 7.9 is not explained. An Euler-characteristic argument would need the full cell structure (edges and higher cells), which the multiplicative theory does not provide (§3.1); a neighbour walk with cells would turn the heuristic into a mass formula and a completeness certificate.

5.2. **The Hermite–Humbert constant.**

Observation 5.3 ([CERTIFIED], 157 fields). A least-squares fit gives

$$\log \gamma_{K,2} \approx 1.172 \log d - 0.614 \log \mathrm{Reg}_K - 1.711 \qquad (R^2 = 0.977,\ \text{residual s.d. } 0.18),$$

against $R^2 = 0.923$ for $\log d$ alone; adding $\log h$ changes nothing (coefficient -0.05). $\gamma_{K,2}/d$ ranges from 0.068 ($D = 46$) to 0.358 ($D = 5$), well below Cohn's $\frac{1}{2}$. The low values of $\gamma_{K,2}/d$ for $D \equiv 1 \pmod 8$ (2 split) are explained by the regulator: the residuals of the fit have mean -0.04 (2 split), -0.03 (2 inert), $+0.02$ (2 ramified).

The regulator enters because the cusp is long: the region of highest points has length Reg_K in the s-direction, so the same volume is spread over a longer, thinner cusp, and the floor of the fundamental domain (whose height is $1/\gamma_{K,2}$) rises. Making this precise gives a lower bound.

Theorem 5.4 ([PROVED] for $h_K = 1$, given Hirzebruch's volume formula)**.**

$$\gamma_{K,2} \;\geq\; \frac{4\pi^2\,\zeta_K(-1)}{\sqrt{d_K}\,\mathrm{Reg}_K}.$$

Proof. Every orbit has a highest point $z \in \mathcal{R}$, and there $G(z) = 1/(y_1y_2) \leq \gamma_{K,2}$, so the set $\mathcal{R}(Y) = \{z \in \mathcal{R} :\ y_1y_2 \geq Y\}$ with $Y = 1/\gamma_{K,2}$ meets every orbit of the group Γ generated by $\mathrm{PGL}_2(\mathcal{O}_K)$ and the Galois swap. Hence its hyperbolic volume is at least that of $\Gamma\backslash\mathbb{H}^2$. The x-part of $\mathcal{R}$ has area $\sqrt{d}/2$ (half a period parallelogram of $\mathcal{O}_K$, of area $\sqrt{d}$). With $p = \log(y_1y_2)$, $q = \log(y_2/y_1)$, $dy_1dy_2/(y_1y_2)^2 = \frac{1}{2}e^{-p}\,dp\,dq$ and $q \in [0, \mathrm{Reg}_K]$, so $\mathrm{vol}\,\mathcal{R}(Y) = \frac{\sqrt{d}}{2} \cdot \frac{\mathrm{Reg}_K}{2} \cdot \frac{1}{Y} = \frac{\sqrt{d}\,\mathrm{Reg}_K\,\gamma_{K,2}}{4}$. On the other hand $[\mathrm{PGL}_2(\mathcal{O}_K) : \mathrm{PSL}_2(\mathcal{O}_K)] = |\mathcal{O}_K^\times/\mathcal{O}_K^{\times 2}| = 4$ and the swap has index 2, so $\mathrm{vol}(\Gamma\backslash\mathbb{H}^2) = 8\pi^2\zeta_K(-1)/8 = \pi^2\zeta_K(-1)$. □

Observation 5.5 ([CERTIFIED]). The bound holds in all 157 fields, including $h > 1$; $\gamma_{K,2}$ divided by the bound ranges over $[1.46, 14.5]$ (minimum at $D = 5$, then $D = 2$, 17), so it is never sharp and the ratio grows with d (the log-log slope of $\gamma_{K,2}$ against $\zeta_K(-1)/(\sqrt{d}\mathrm{Reg}_K)$ is 1.27).

Remark 5.6. Since $\zeta_K(-1) \asymp d^{3/2}L(2,\chi_d)$ with $L(2,\chi_d) \in [\zeta(4)/\zeta(2), \zeta(2)]$, the theorem gives $\gamma_{K,2} \gg d/\mathrm{Reg}_K$, and $\mathrm{Reg}_K \ll \sqrt{d}\log d$, so $\gamma_{K,2} \gg \sqrt{d}/\log d$ unconditionally (for $h = 1$). Together with Cohn's $\gamma_{K,2} \leq d/2$ this brackets the growth; the data sit near $d^{1.17}\mathrm{Reg}_K^{-0.61}$. Questions: is $\gamma_{K,2} \asymp d/\mathrm{Reg}_K$ (the data say no: exponents 1.17, −0.61)? Is $\limsup \gamma_{K,2}/d < \frac{1}{2}$? The extension of the proof to $h > 1$ needs the volume of the cusp regions of all charts; the data say the bound still holds.

5.3. **Minimal vectors and automorphisms.**

Observation 5.7 ([CERTIFIED]). Of the 55444 perfect classes, 44643 (80.5%) have exactly 5 minimal classes, the minimum for perfection; 8472 have 6; the maximum is 24 ($D = 263$). The stabiliser in $\mathrm{GL}_2(\mathcal{O}_K)/\mathcal{O}_K^\times$ is trivial for 53228 classes, of order 2 for 2106, and of order $3, 4, 6, 10, 24$ for $4, 64, 40, 1, 1$ classes.

Observation 5.8 ([CERTIFIED], 157 fields). All 45 classes whose stabiliser has order divisible by 3 have $\frac{3}{4}\gamma$ equal to an integer of the form $a^2 + ab + b^2$ (a norm from $\mathbb{Z}[\zeta_3]$). Of the 64 classes with stabiliser of order 4, 51 have γ an integer of the form $a^2 + b^2$. The stabiliser of order 10 is that of the unique class of $\mathbb{Q}(\sqrt{5})$ ($\gamma = 4/\sqrt{5}$), and the one of order 24 that of $\mathbb{Q}(\sqrt{3})$ ($\gamma = 4$).

Proposition 5.9 ([PROVED])**.** *Let $g \in \mathrm{GL}_2(\mathcal{O}_K)$ stabilise a form and have order 3 in $\mathrm{PGL}_2(K)$. Then $g^2 - tg + \delta = 0$ with $t \in \mathcal{O}_K$, $\delta = \det g = t^2$, and $\mathcal{O}_K^2$ is a module over the order $\mathcal{O}_K[g] \cong \mathcal{O}_K[\frac{1}{2}t(1+\sqrt{-3})]$ of the CM field $L = K(\sqrt{-3})$.*

Proof. g^3 is scalar and g is not, so the eigenvalues are $\lambda, \lambda\zeta_3$; then $t = \lambda(1+\zeta_3) = -\lambda\zeta_3^2$ and $\delta = \lambda^2\zeta_3$, so $t^2 = \delta$ and the discriminant is $t^2 - 4\delta = -3t^2$; $t \neq 0$ since δ is a unit, so $g = \frac{t}{2}(1+\sqrt{-3})$ in $L = K(\sqrt{-3})$. □

Heuristically, a form fixed by g is, in each embedding, invariant under a rotation of order 3, hence a multiple of the hexagonal form, and $S[v]$ becomes a norm form of L on a rank-one $\mathcal{O}_K[g]$-module. Since $L = K(\sqrt{-3}) \supset \mathbb{Q}(\sqrt{-3})$, norms from L to $\mathbb{Q}$ factor through $\mathbb{Q}(\zeta_3)$; we expect this to explain the shape $\frac{4}{3}(a^2 + ab + b^2)$ in general [CONJECTURE] (note $\frac{4}{3} = \gamma_2^2$, the square of the Hermite constant of the hexagonal lattice). The same mechanism with ζ_4, ζ_5 ($\sqrt{5} \in K$) and ζ_{12} ($\sqrt{3} \in K$) accounts for the other symmetric forms. This is the correspondence with quadratic (CM) extensions of K asked about: forms with extra automorphisms carry the structure of hermitian forms over CM extensions L/K (Proposition 5.9 for order 3), and we expect their γ to be governed by norms from L [CONJECTURE].

5.4. **Eutaxy.** The median fraction of extreme classes per field is 0.33, against 0.97 for perfect unary forms over real quadratic fields (`perfect_forms`, FINDINGS §5); weakly eutactic classes occur (80, in 52 fields), unlike the unary case. The number of extreme classes grows more slowly than n_{perf} (slope 0.82 against $\zeta_K(-1)$).

6. Next steps

6.1. **Completing $D \le 300$ and certifying the lists.** The 25 fields with $D \le 300$ whose $\gamma_{K,2}$ is not certified (large units) are the obvious gap; the enumeration itself would run on them (it does not need $\gamma_{K,2}$, only Cohn's bound), but its cost grows with the length of the cusp. Certified completeness needs the interval-Newton version of §3.5 (estimated cost: a small constant factor, since most leaves have no solution and are rejected after one interval evaluation).

6.2. **Ternary forms over real quadratic fields.** The space of ternary forms modulo the two scalings is $(\mathrm{SL}_3(\mathbb{R})/\mathrm{SO}_3)^2$, of dimension 10. A box enumeration in dimension 10 is not feasible: the number of leaves scales like $w_{\min}^{-10}$ times the volume of $\mathrm{PGL}_3(\mathcal{O}_K)\backslash(\mathrm{SL}_3(\mathbb{R})/\mathrm{SO}_3)^2$, which already for $\mathbb{Q}(\sqrt{5})$ is far beyond the binary case. Feasible routes: (a) a multiplicative Voronoi walk (§3.5, item 6), whose cost is proportional to the output; (b) a search for $\gamma_{K,3}$ alone, by local maximisation from many seeds together with structured candidates (the lattices whose trace form is similar to E_6, or hermitian structures over CM extensions as in Proposition 5.9), with a branch-and-bound certificate only if a good candidate is found; (c) in the additive normalisation the Voronoi algorithm works in any rank, and `perfect_forms` is the place for it. We would start with (b) for $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{3})$, $\mathbb{Q}(\sqrt{5})$.

6.3. **Binary forms over totally real fields of higher degree.** For a totally real cubic field the space of binary forms modulo the three scalings is $\mathbb{H}^3$ (dimension 6), $\gamma_{K,2}$ is again the reciprocal height of the floor, and every step of §3 generalises (products of three factors, ε in a rank-2 unit lattice, 6-subsets for Newton). The cost of the $\gamma_{K,2}$-certificate and of the enumeration is proportional to the volume ($\asymp \zeta_K(-1)$ again) times $w_{\min}^{-6}$ instead of $w_{\min}^{-4}$: feasible for the smallest cubic fields ($d = 49, 81, 148, 169, \dots$), out of reach for large ones without the improvements above. The code is written for two embeddings throughout; a degree-n version is a rewrite of `humbert.py`, `certify.py` and `perfect.py`, not a patch.

6.4. **Further questions suggested by the data.**

(1) Prove $n_{\mathrm{perf}} \asymp \zeta_K(-1)$ (Conjecture 5.2), perhaps first a lower bound by a covering argument as in Theorem 5.4.
(2) Explain $\gamma_{K,2} \approx 0.18\, d^{1.17}\mathrm{Reg}_K^{-0.61}$; extend Theorem 5.4 to $h > 1$.
(3) Classify the symmetric forms by CM extensions L/K (which L, which orders, which ideal classes) and compute their γ from norms in L.
(4) Compare the multiplicative and additive extreme forms field by field: `perfect_forms` finds $\gamma_{\mathrm{add}}^4/(16\gamma_{\mathrm{mult}}^2) \in [1.56, 46.1]$, never 1.

7. Literature to check

We cite only what is in the project files or in the verified bibliography of `perfect_forms`. To place the results we would like the following (PDF or full reference):

(1) R. Coulangeon, M. I. Icaza, M. O'Ryan, the 2007 paper on extreme forms over number fields ("CIO07" in our notes; uploaded earlier but no longer in the sandbox) **[REF?]**.
(2) M. Wagner's thesis and any later Pohst–Wagner papers on $\gamma_{K,2}$ **[REF?]**.
(3) M. I. Icaza's papers defining the multiplicative constant and its bounds (Hermite constant and extreme forms for algebraic number fields) **[REF?]**; R. Baeza and M. I. Icaza on Humbert's constant **[REF?]**.
(4) J. L. Thunder and/or T. Watanabe, K. Ohno on bounds for Hermite constants of number fields (lower bounds of Minkowski–Hlawka type) **[REF?]**.

(5) F. Hirzebruch, Hilbert modular surfaces, and G. van der Geer's book of the same title, for the volume and Euler characteristic of $\mathrm{PSL}_2(\mathcal{O}_K)\backslash\mathbb{H}^2$, Siegel's formula for $\zeta_K(-1)$, and the cusp geometry **[REF?]**.
(6) Work on hermitian forms over CM fields (perfect and extreme hermitian forms, e.g. Coulangeon's and Okuda–Yano's generalisations of Voronoi's theorem [7]) for Proposition 5.9.
(7) P. E. Gunnells and D. Yasaki [4] and A. Leibak [6] for the additive side, already in hand.

Appendix A. Counts per field

Table 2: Per field: D, h_K, $\gamma_{K,2}$ (rounded), number of perfect and of extreme classes [CERTIFIED].

D	h	$\gamma_{K,2}$	perf	ext	D	h	$\gamma_{K,2}$	perf	ext	D	h	$\gamma_{K,2}$	perf	ext
2	1	2.106	1	1	87	2	76	210	52	187	2	105.4	810	207
3	1	4	1	1	89	1	10.45	28	10	190	2	100.8	619	171
5	1	1.789	1	1	91	2	33.98	210	75	194	2	146.1	721	228
6	1	5	3	3	93	1	20.27	17	7	195	4	190.9	658	139
7	1	4.252	4	4	95	2	101.3	241	72	197	1	38.56	67	31
10	2	8.706	7	5	97	1	9.204	31	19	202	2	119.1	832	164
11	1	9.333	8	7	101	1	20.43	31	12	203	2	157	671	170
13	1	4.035	2	2	102	2	65.78	266	87	205	2	48.08	88	30
14	1	11.11	11	6	103	1	43.52	378	119	206	1	149.1	886	220
15	2	12.55	13	11	105	2	25.6	36	10	209	1	17.33	95	38
17	1	2.608	2	2	106	2	49.65	303	107	210	4	161.4	714	226
19	1	8.056	16	9	107	1	89.33	291	99	213	1	44.95	72	39
21	1	5.333	2	2	109	1	9.542	26	17	215	2	193.3	819	183
22	1	10.8	23	16	110	2	90.91	238	70	218	2	129.9	729	184
23	1	17.33	25	16	111	2	64.51	340	113	219	4	196	899	208
26	2	23.55	36	18	113	1	12.07	43	23	221	2	57.78	108	52
29	1	5.792	3	3	114	2	82.67	358	97	222	2	161.4	822	220
30	2	28	38	17	115	2	101.3	303	69	223	3	148.7	906	245
31	1	11.31	37	17	118	1	50.13	431	120	226	8	207.7	976	280
33	1	4.571	3	3	119	2	100.8	373	111	227	1	169.3	801	248
34	2	34.01	78	38	122	2	129.4	301	60	229	3	50.7	95	33
35	2	33.12	41	15	123	2	81.73	368	104	230	2	241.3	844	160
37	1	5.736	4	4	129	1	12.31	51	21	231	4	218.7	1032	263
38	1	30.16	34	19	130	4	117.6	439	114	233	1	21.55	143	54
39	2	25	54	31	131	1	91.24	465	114	235	6	281.3	1037	250
41	1	5.321	10	5	133	1	25.35	49	33	237	1	52.05	98	36
42	2	36.63	70	27	134	1	91.66	524	135	238	2	137.5	1357	280
43	1	21.13	87	27	137	1	13.29	47	22	246	2	90.05	1091	363
46	1	12.5	84	39	138	2	103.4	365	93	247	2	106.6	1181	240
47	1	33.41	73	38	141	1	28.09	63	27	253	1	29.8	129	43
51	2	36.6	91	44	142	3	106.2	448	137	254	3	184	1075	322
53	1	12.42	9	1	143	2	138.4	339	76	255	4	244.1	1058	257
55	2	40.11	118	48	145	4	35.14	102	36	257	3	52.21	145	47
57	1	9.333	10	5	146	2	129.3	520	165	258	2	204.3	1074	262
58	2	40.1	123	34	149	1	26.8	50	14	263	1	179.5	1144	303
59	1	44.1	110	45	154	2	78.99	534	153	265	2	34.11	246	78
61	1	6.456	8	7	155	2	125.8	531	118	266	2	146.7	1012	250
62	1	52.08	111	47	157	1	20.82	62	26	267	2	133.2	1124	320
65	2	17.07	21	7	158	1	109.1	578	142	269	1	49.51	126	43
66	2	52	142	47	159	2	111.2	565	164	273	2	33.33	267	113
67	1	29.27	177	65	161	1	12.63	79	32	274	4	209.6	1467	273
69	1	13.16	15	7	165	2	41.72	52	21	277	1	34.4	151	67
70	2	52.23	133	31	167	1	129.3	477	152	278	1	201	1095	296
71	1	50.08	210	67	170	4	158.8	566	170	282	2	183.6	1157	276

Table 2: Per field: D, h_K, $\gamma_{K,2}$ (rounded), number of perfect and of extreme classes [CERTIFIED].

D	h	$\gamma_{K,2}$	perf	ext	D	h	$\gamma_{K,2}$	perf	ext	D	h	$\gamma_{K,2}$	perf	ext
73	1	4.977	17	15	173	1	35.02	48	29	285	2	66.22	122	61
74	2	47.85	143	37	174	2	124	523	176	287	2	199.4	1134	358
77	1	17.63	12	8	177	1	14.3	88	40	290	4	279.4	1125	289
78	2	57.48	162	50	178	2	110	760	217	291	4	208.6	1427	398
79	3	52.14	203	74	181	1	17.92	68	35	293	1	70.92	109	39
82	4	75.75	228	57	182	2	176.7	539	111	298	2	165.3	1431	349
83	1	62.3	181	64	183	2	109	611	177	299	2	198.5	1223	304
85	2	22.1	47	14	185	2	36.13	106	16					
86	1	58.86	271	84	186	2	101	647	139					

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